D3V3NDRA CLASSES

INTERNATIONAL MATHEMATICS OLYMPIAD (IMO) • TRAINING SERIES

CHAPTER 10 • OPERATIONS WITH INTEGERS

Grade VII (Class 7) • NCERT Advanced Curriculum Benchmark • Total Questions: 50

Candidate Name: ..... **Roll No:** ..... **School / Center:** ..... **Date:** .....

Time: 60 Mins | **Max Marks:** 100 **Total Questions:** 50 MCQs | **Marking:** +2 / -0 (No Negative) **Structure:** Sec 1 (15 LR) • Sec 2 (20 MR) • Sec 3 (10 EM) • Sec 4 (5 Achievers)

SECTION 1: LOGICAL REASONING (Q1 – Q15)

Q1. Consider the geometric sequence of alternating integers:

$$-2,\ 4,\ -8,\ 16,\ -32,\ 64,\ \ldots$$

By analyzing the common multiplier connecting consecutive terms, what is the SUM of the 7th and 8th terms?

(A) 128 **(B)** 384
(C) -128 **(D)** -384

Q2. In a 3×3 magic square with integer entries, the sum of numbers along every row, column, and diagonal is strictly equal to -15. If the center cell contains -5 and the top-left cell contains -8, what integer must occupy the bottom-right cell?

(A) -7 **(B)** -4
(C) 2 **(D)** -2

Q3. Three non-zero integers a, b, c satisfy the sign relations: $a \times b < 0$, $b \times c > 0$, and $a > 0$. Which of the following statements is GUARANTEED to be mathematically true?

(A) $a + c > 0$ **(B)** $b \times c < 0$
(C) $a \times c < 0$ **(D)** $b > 0$

Q4. An operation $\otimes$ on integers is defined by the algebraic rule:

$$a \otimes b = a \times b - (a + b) + 1$$

Evaluate the numerical value of $(-3) \otimes (-5)$.

(A) 16 **(B)** 24
(C) -24 **(D)** 8

Q5. An insect starts at position -12 on an integer number line. It makes 5 consecutive jumps of $+7$ units each, followed by 4 consecutive jumps of -9 units each. What is the insect's final position on the number line?

(A) -13 **(B)** -11
(C) 13 **(D)** -1

Q6. Four mathematical products are evaluated:

$P_1 = (-1)^1 \times (-2)^2, \quad P_2 = (-3)^3 \times (-1)^4, \quad P_3 = (-2)^5 \times (-1)^1, \quad P_4 = (-1)^{10} \times (-4)^3$

Which of the four products is strictly POSITIVE while all others are negative?

(A) P_2 **(B)** P_4
(C) P_3 **(D)** P_1

Q7. On a balanced numerical scale, 3 identical unknown weight tokens x combined with a negative counter-weight of -14 exactly balance a net value of -2. What is the integer value of x?

(A) -6 **(B)** -4
(C) 4 **(D)** 6

Q8. Four unknown integers p, q, r, s satisfy the equality chain:

$$p - 6 = q + 4 = r - 1 = s + 9$$

Which of the following lists the four integers in strictly ASCENDING order (from smallest to largest)?

(A) $q < s < r < p$ **(B)** $s < q < r < p$
(C) $p < r < q < s$ **(D)** $s < r < q < p$

Q9. In a 2×2 integer matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the cross-difference value is defined as $(ad - bc)$. If $a = -4, d = -6, b = 5$, and the cross-difference value is 4, what is the integer value of c?

(A) 4 **(B)** -4
(C) 5 **(D)** -5

Q10. If the integer cryptarithm $(-A) \times B = -72$ and $A - B = -1$ holds true, where A and B are positive single-digit integers, what is the value of $(A + B)$?

(A) 15 **(B)** 19
(C) 17 **(D)** 18

Q11. A marker on a vertical integer line starts at 0 and performs 50 sequential moves: Move 1: $+1$, Move 2: -2, Move 3: $+3$, Move 4: -4, . . ., Move 50: -50. What is the marker's final coordinate?

(A) -25 **(B)** 0
(C) -50 **(D)** 25

Q12. Which of the following expressions evaluates to a strictly NEGATIVE integer?

(A) $-(-5)^2 \times (-1)^3$ **(B)** $(-2) \times (-4) \times 0$
(C) $(-3)^2 \times (-2)^2$ **(D)** $(-1)^{101} \times (-2)^4$

Q13. Consider the sequence:

$$5,\ -10,\ -5,\ 10,\ 15,\ -30,\ -15,\ 30,\ [\ ?\ ]$$

By identifying the repeating operational logic, what is the next integer term in the sequence?

(A) 60 **(B)** -35
(C) -60 **(D)** 35

Q14. How many distinct integer values of x satisfy the strict quadratic inequality:

$$(x - 3)(x + 5) < 0$$

(A) 8 **(B)** 6
(C) 9 **(D)** 7

Q15. If a is a negative integer and b is a positive integer, which of the following expressions is GUARANTEED to be strictly NEGATIVE?

(A) $a - b$ **(B)** $(-a) \times b$
(C) $b - a$ **(D)** $b + a$

SECTION 2: MATHEMATICAL REASONING (Q16 – Q35)

Q16. Evaluate the multi-step integer expression with nested brackets:

$$(-15) + [(-8) \times (-4) - \{(-24) \div (-3) + (-5)\}] \times (-2)$$

(A) 43 **(B)** 73
(C) -43 **(D)** -73

Q17. Evaluate the arithmetic product smartly using the distributive property:

$$(-198) \times 47 + (-198) \times 53$$

(A) $-19,000$ **(B)** $-19,800$
(C) $19,800$ **(D)** $-1,980$

Q18. What is the SUM of all integer solutions to the absolute value equation:

$$|2x - 7| = 15$$

(A) 15 **(B)** 7
(C) 11 **(D)** -4

Q19. The product of three consecutive integers is -120. What is the SUM of these three consecutive integers?

(A) -15 **(B)** -18
(C) -12 **(D)** 15

Q20. Evaluate the finite sum of alternating powers of -1:

$$S = (-1)^1 + (-1)^2 + (-1)^3 + \cdots + (-1)^{2026}$$

(A) 1013 **(B)** 1
(C) 0 **(D)** -1

Q21. According to Euclid's division lemma for integers, when $a = -47$ is divided by $b = 5$, it is written as $a = bq + r$ with $0 \leq r < b$. What are the unique integer quotient q and remainder r?

(A) $q = -9,\ r = -2$ **(B)** $q = -10,\ r = -3$
(C) $q = -10,\ r = 3$ **(D)** $q = -9,\ r = 2$

Q22. If $x = -3, y = -2$, and $z = 4$, evaluate the numerical value of the algebraic fraction:

$$\frac{x^2 - 2yz}{y^3 + xz}$$

(A) 1 **(B)** $-\frac{25}{4}$
(C) $\frac{5}{4}$ **(D)** $-\frac{5}{4}$

Q23. How many integers n satisfy the compound inequality:

$$-18 \leq 3n - 4 < 23$$

(A) 14 **(B)** 12
(C) 13 **(D)** 15

Q24. Evaluate using the distributive property:

$$125 \times (-37) - 125 \times 63$$

(A) $-12,500$ **(B)** $-3,250$
(C) $12,500$ **(D)** $-1,250$

Q25. A fallacious mathematical argument is written as:
Step 1: $(-1)^3 = -1$
Step 2: $(-1)^{6/2} = -1$
Step 3: $[(-1)^6]^{1/2} = 1^{1/2} = 1$
Step 4: Therefore, $-1 = 1$.
In which step does the fundamental algebraic error occur?

(A) Step 3 **(B)** Step 2
(C) Step 4 **(D)** Step 1

Q26. Evaluate the telescoping sum of 200 consecutive integers with alternating signs:

$$S = (1 - 2) + (3 - 4) + (5 - 6) + \cdots + (199 - 200)$$

(A) -50 **(B)** -100
(C) 100 **(D)** 0

Q27. If a and b are integers such that $a \times b = -36$, what is the ABSOLUTE MINIMUM possible value of the sum $(a + b)$?

(A) -35 **(B)** -16
(C) -37 **(D)** -12

Q28. If the integer variables x and y are bounded by $-8 \leq x \leq 5$ and $-12 \leq y \leq -3$, what is the MAXIMUM possible value of $(x - y)$?

(A) 14 **(B)** 17
(C) 11 **(D)** 8

Q29. Which of the following integers is its own multiplicative inverse (reciprocal) in the set of all integers?

(A) 2 **(B)** -1
(C) -2 **(D)** 0

Q30. Evaluate the alternating integer sum:

$$100 - 99 + 98 - 97 + 96 - 95 + \cdots + 2 - 1$$

(A) 51 **(B)** 50
(C) 0 **(D)** 100

Q31. Solve for the integer x in the linear equation:

$$-4(3x - 5) + 6 = -2(5x + 4)$$

(A) -17 **(B)** 13
(C) -13 **(D)** 17

Q32. If the side lengths of a triangle are strictly integers of lengths $5, 11$, and x, how many distinct integer values can x assume?

(A) 11 **(B)** 9
(C) 10 **(D)** 8

Q33. If k is an odd negative integer, what is the mathematical sign of the product $(-k)^3 \times k^4$?

(A) Strictly negative **(B)** Cannot be determined without$k$
(C) Zero **(D)** Strictly positive

Q34. Evaluate the arithmetic expression involving absolute values:

$$|-12-(-7)|+|5-(-8)|-|-9|$$

(A) 11 **(B)** 9
(C) 7 **(D)** 13

Q35. Evaluate the continuous product of signed integers:

$$(-2)\times(-3)\times(-4)\times(-5)\times(-1)$$

(A) -60 **(B)** -120
(C) 60 **(D)** 120

SECTION 3: EVERYDAY MATHEMATICS (Q36 – Q45)

Q36. A merchant begins the day with an active bank account balance of $\text{Rs}\ 4,500$. During the business day, he executes three consecutive debit payments of $\text{Rs}\ 2,800$, $\text{Rs}\ 3,200$, and $\text{Rs}\ 1,500$. What is his net bank account balance at the end of the day?

(A) $-\text{Rs}\ 1,500$ **(B)** $\text{Rs}\ 3,000$
(C) $-\text{Rs}\ 3,000$ **(D)** $-\text{Rs}\ 2,500$

Q37. A submarine patrols at a depth of 380 metres below sea level (elevation -380 m). Directly overhead, an anti-submarine reconnaissance aircraft flies at an altitude of $1,420$ metres above sea level ($+1420$ m). What is the vertical distance separating the aircraft and the submarine?

(A) $1,040$ metres **(B)** $1,780$ metres
(C) $1,800$ metres **(D)** $1,820$ metres

Q38. At 6:00 AM, the temperature at a Himalayan meteorological station was -4°C. By noon, the temperature had risen by 9°C, but by midnight, a cold front caused it to drop by 13°C. What was the temperature recorded at midnight?

(A) -8°C **(B)** 8°C
(C) -6°C **(D)** -10°C

Q39. In a 25-question competitive mathematics test, $+4$ marks are awarded for each correct answer, -2 marks are deducted for each incorrect answer, and 0 marks are given for unattempted questions. Arjun attempted 22 questions, out of which 16 were correct. What was his final examination score?

(A) 48 marks **(B)** 52 marks
(C) 58 marks **(D)** 64 marks

Q40. A mining elevator begins descending into a vertical shaft from an access platform located 18 metres ABOVE ground level ($+18$ m). It descends at a constant speed of 6 metres per minute. What is its elevation relative to ground level after 45 minutes of descent?

(A) -288 metres **(B)** -234 metres
(C) -270 metres **(D)** -252 metres

Q41. Over a 5-month period, an enterprise recorded monthly financial performance as follows: Month 1: $+\text{Rs}\ 45,000$; Month 2: $-\text{Rs}\ 28,000$; Month 3: $-\text{Rs}\ 35,000$; Month 4: $+\text{Rs}\ 52,000$; Month 5: $-\text{Rs}\ 14,000$. What was the aggregate profit or loss over this period?

(A) Net profit of$\text{Rs}\ 20,000$ **(B)** Net loss of$\text{Rs}\ 20,000$
(C) Net profit of$\text{Rs}\ 18,000$ **(D)** Net loss of$\text{Rs}\ 14,000$

Q42. The shore of the Dead Sea is at an elevation of -430 metres relative to sea level. The shore of Lake Assal is at an elevation of -155 metres. By how many metres is Lake Assal higher in elevation than the Dead Sea?

(A) 275 metres **(B)** 585 metres
(C) -275 metres **(D)** 285 metres

Q43. Over four consecutive trading sessions, a stock index recorded day-to-day point changes of -145 points, $+280$ points, -95 points, and -110 points. What was the net point change of the index over the four days?

(A) -60 points **(B)** $+70$ points
(C) $+110$ points **(D)** -70 points

Q44. A skyscraper has 35 floors above ground (Floors 1 to 35) and 4 basement levels (Levels -1 to -4), with ground level designated as 0. If an elevator starts at Basement Level -3 and ascends 19 levels, which floor does it arrive at?

(A) Floor15 **(B)** Floor19
(C) Floor16 **(D)** Floor17

Q45. A chemical solution initially at $+32^\circ$C is placed inside a cryogenic cooling chamber that decreases its temperature at a uniform rate of 4°C every 5 minutes. What will be the temperature of the solution after 1 hour?

(A) -18°C **(B)** 16°C
(C) -16°C **(D)** -12°C

SECTION 4: ACHIEVERS SECTION / HOTS (Q46 – Q50)

Q46. What is the ABSOLUTE MINIMUM possible numerical value achievable by the piecewise absolute value function:

$$E(x)=|x-2|+|x-5|+|x-11|$$

where x is an integer?

(A) 9 **(B)** 7
(C) 8 **(D)** 11

Q47. How many ordered pairs of integers (x, y) satisfy the Diophantine integer equation:

$$xy+3x-5y=33$$

(A) 6 **(B)** 16
(C) 12 **(D)** 8

Q48. Twenty integers, each either $+1$ or -1, are arranged around a circle. The product of each adjacent pair is calculated, generating 20 products $p_1, p_2, \ldots, p_{20}$. If the sum of these 20 products is strictly equal to 0, how many of the 20 products must equal -1?

(A) 8 **(B)** 10
(C) 12 **(D)** Such an arrangement is impossible

Q49. Four distinct integers a, b, c, d are chosen from the set $\{-5, -3, -1, 2, 4, 6\}$. What is the ABSOLUTE MAXIMUM possible value achievable by the product:

$$(a-b)\times(c-d)$$

(A) 88 **(B)** 66
(C) 72 **(D)** 77

Q50. Evaluate the net sum of the 101-term alternating integer series:

$$S=1-2+3-4+5-6+\cdots+99-100+101$$

(A) -50 **(B)** 50
(C) -51 **(D)** 51

ANSWER KEY & DETAILED STEP-BY-STEP SOLUTIONS

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QUICK ANSWER KEY MATRIX (50 MCQS)

1 **A**	2 **D**	3 **C**	4 **B**	5 **A**	6 **C**	7 **C**	8 **B**	9 **A**	10 **C**
11 **A**	12 **D**	13 **D**	14 **D**	15 **A**	16 **D**	17 **B**	18 **B**	19 **A**	20 **C**
21 **C**	22 **D**	23 **C**	24 **A**	25 **A**	26 **B**	27 **A**	28 **B**	29 **B**	30 **B**
31 **D**	32 **B**	33 **D**	34 **B**	35 **B**	36 **C**	37 **C**	38 **A**	39 **B**	40 **D**
41 **A**	42 **A**	43 **D**	44 **C**	45 **C**	46 **A**	47 **C**	48 **B**	49 **D**	50 **D**

COMPREHENSIVE STEP-BY-STEP SOLUTIONS & CONCEPTUAL FOUNDATIONS

Q1 [LOGICAL REASONING] **Correct Option: (A)** 128

Geometric Alternating Sequence:
The sequence has first term $T_1 = -2$ and common ratio $r = -2$:
• $T_7 = (-2)^7 = -128$
• $T_8 = (-2)^8 = +256$
Sum of the 7th and 8th terms:
$$T_7 + T_8 = (-128) + 256 = 128$$

Q2 [LOGICAL REASONING] **Correct Option: (D)** -2

Magic Square Invariant:
The main diagonal containing top-left (TL), center (C), and bottom-right (BR) must sum to the constant sum $S = -15$:
$$TL + C + BR = -15$$
Substituting $TL = -8$ and $C = -5$:
$$(-8) + (-5) + BR = -15 \implies -13 + BR = -15 \implies BR = -15 - (-13) = -2$$

Q3 [LOGICAL REASONING] **Correct Option: (C)** $a \times c < 0$

Rules of Signs Deduction:
1. Given $a > 0$ (positive) and $a \times b < 0$: since the product is negative, b must be negative ($b < 0$).
2. Given $b \times c > 0$: since $b < 0$, for their product to be positive, c must also be negative ($c < 0$).
3. Since $a > 0$ and $c < 0$, their product must be strictly negative:
$$a \times c < 0$$

Q4 [LOGICAL REASONING] **Correct Option: (B)** 24

Relational Operation on Integers:
1. Product term: $a \times b = (-3) \times (-5) = +15$.
2. Sum term: $a + b = (-3) + (-5) = -8$.
3. Combine:
$$(-3) \otimes (-5) = 15 - (-8) + 1 = 15 + 8 + 1 = 24$$

Q5 [LOGICAL REASONING] **Correct Option: (A)** -13

Number Line Displacement:
$$\text{Final Position} = -12 + 5(+7) + 4(-9) = -12 + 35 - 36 = -13$$

Q6 [LOGICAL REASONING] **Correct Option: (C)** P_3

Sign Rules for Powers:
• $P_1 = (-1) \times 4 = -4$ (Negative)
• $P_2 = (-27) \times 1 = -27$ (Negative)
• $P_3 = (-32) \times (-1) = +32$ (Positive)
• $P_4 = 1 \times (-64) = -64$ (Negative)
Therefore, P_3 is the only positive product.

Q7 [LOGICAL REASONING] **Correct Option: (C)** 4

Balance Scale Equation:
$$3x + (-14) = -2 \implies 3x - 14 = -2 \implies 3x = -2 + 14 = 12 \implies x = 4$$

Q8 [LOGICAL REASONING] **Correct Option: (B)** $s < q < r < p$

Relative Magnitude Logic:
Let the common constant value of the chain be K:
• $p = K + 6$
• $q = K - 4$
• $r = K + 1$
• $s = K - 9$
Comparing the values: $K - 9 < K - 4 < K + 1 < K + 6$.
Therefore, the ascending order is $s < q < r < p$.

Q9 [LOGICAL REASONING] **Correct Option: (A)** 4

Matrix Cross-Difference:
$$ad - bc = (-4)(-6) - 5c = 24 - 5c$$
Given $24 - 5c = 4$:
$$5c = 24 - 4 = 20 \implies c = 4$$

Q10 [LOGICAL REASONING] **Correct Option: (C)** 17

Cryptarithm Deduction:
1. $(-A) \times B = -72 \implies A \times B = 72$.
2. $A - B = -1 \implies B - A = 1$, meaning A and B are consecutive integers.
3. Since $8 \times 9 = 72$ and $9 - 8 = 1$, we have $A = 8$ and $B = 9$.
$$A + B = 8 + 9 = 17$$

Q11 [LOGICAL REASONING] **Correct Option: (A)** -25

Pairwise Cancellation Pattern:
Group the 50 moves into 25 consecutive pairs:
$$\text{Final Coordinate} = (1 - 2) + (3 - 4) + (5 - 6) + \cdots + (49 - 50)$$
Each of the 25 pairs evaluates to -1:
$$25 \times (-1) = -25$$

Q12 [LOGICAL REASONING] **Correct Option: (D)** $(-1)^{101} \times (-2)^4$

Evaluating Signs:
• Option A: $(-1)^{101} \times (-2)^4 = (-1) \times 16 = -16$ (Negative)
• Option B: $(-3)^2 \times (-2)^2 = 9 \times 4 = 36$ (Positive)
• Option C: $-(-5)^2 \times (-1)^3 = -25 \times (-1) = +25$ (Positive)
• Option D: Evaluates to 0 (Neither positive nor negative).

Q13 [LOGICAL REASONING] **Correct Option: (D)** 35

Two-Step Alternating Operations:
Examine the operations between consecutive numbers:
• $5 \times (-2) = -10$
• $-10 + 5 = -5$
• $-5 \times (-2) = 10$
• $10 + 5 = 15$
• $15 \times (-2) = -30$
• $-30 + 15 = -15$
• $-15 \times (-2) = 30$
The alternating pattern is: multiply by -2, then add $+5$ (or step increment).
Following the pattern, after multiplying by -2 to get 30, we add $+5$:
$$30 + 5 = 35$$

Q14 [LOGICAL REASONING] **Correct Option: (D)** 7

Sign Analysis of Factors:
For the product $(x - 3)(x + 5)$ to be strictly negative, one factor must be positive and the other negative:
This occurs strictly when:
$$-5 < x < 3$$
The integers in this open interval are:
$$x \in \{-4, -3, -2, -1, 0, 1, 2\}$$
Total count $= 2 - (-4) + 1 = 7$ integers.

Q15 [LOGICAL REASONING] **Correct Option: (A)** $a - b$

Negative Minus Positive Property:
Given $a < 0$ and $b > 0$:
$$a - b = a + (-b)$$
Since both a and $(-b)$ are strictly negative, their sum $a - b$ is strictly negative.
(Note: $b - a = b + (-a)$ is positive plus positive, hence strictly positive).

Q16 [MATHEMATICAL REASONING] **Correct Option: (D)** -73

BODMAS Order of Operations:
1. Innermost braces: $(-24) \div (-3) = 8$.
Then $8 + (-5) = 3$.
2. Multiplication inside brackets: $(-8) \times (-4) = 32$.
3. Inside brackets: $32 - 3 = 29$.
4. Multiplication: $29 \times (-2) = -58$.
5. Addition:

$$(-15) + (-58) = -73$$

Q17 [MATHEMATICAL REASONING] **Correct Option: (B)** $-19,800$

Distributive Property of Multiplication:
Factor out (-198):

$$(-198) \times (47 + 53) = (-198) \times 100 = -19,800$$

Q18 [MATHEMATICAL REASONING] **Correct Option: (B)** 7

Absolute Value Solutions:
Case 1: $2x - 7 = 15 \implies 2x = 22 \implies x = 11$.
Case 2: $2x - 7 = -15 \implies 2x = -8 \implies x = -4$.
Sum of the solutions:

$$11 + (-4) = 7$$

Q19 [MATHEMATICAL REASONING] **Correct Option: (A)** -15

Consecutive Integer Product:
Let the integers be $(n-1), n, (n+1)$.
Since their product is negative, test negative integers:

$$(-6) \times (-5) \times (-4) = 30 \times (-4) = -120$$

The three consecutive integers are $-6, -5, -4$.
Their sum is:

$$(-6) + (-5) + (-4) = -15$$

Q20 [MATHEMATICAL REASONING] **Correct Option: (C)** 0

Alternating Sum Invariant:
Each consecutive pair $(-1)^{2k-1} + (-1)^{2k} = (-1) + 1 = 0$.
Since there are 2026 terms (an even number of terms), they pair up into exactly 1013 pairs of 0:

$$S = 1013 \times 0 = 0$$

Q21 [MATHEMATICAL REASONING] **Correct Option: (C)** $q = -10,\ r = 3$

Integer Division with Non-Negative Remainder:
The remainder r must strictly satisfy $0 \le r < 5$.
If we choose $q = -9$: $5 \times (-9) = -45$, so remainder would be -2 (invalid, since remainder cannot be negative).
We must choose $q = -10$:

$$5 \times (-10) + 3 = -50 + 3 = -47$$

Here $r = 3$ satisfies $0 \le 3 < 5$. Thus $q = -10, r = 3$.

Q22 [MATHEMATICAL REASONING] **Correct Option: (D)** $-\frac{5}{4}$

Evaluation with Signed Integers:
1. Numerator:

$$x^2 - 2yz = (-3)^2 - 2(-2)(4) = 9 - (-16) = 9 + 16 = 25$$

2. Denominator:

$$y^3 + xz = (-2)^3 + (-3)(4) = -8 + (-12) = -20$$

3. Fraction:

$$\frac{25}{-20} = -\frac{5}{4}$$

Q23 [MATHEMATICAL REASONING] **Correct Option: (C)** 13

Linear Integer Inequality:
Add 4 across the inequality:

$$-18 + 4 \le 3n < 23 + 4 \implies -14 \le 3n < 27$$

Divide by 3:

$$-\frac{14}{3} \le n < 9 \implies -4.67 \le n < 9$$

The integer values are $n \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8\}$.
Count $= 8 - (-4) + 1 = 13$ integers.

Q24 [MATHEMATICAL REASONING] **Correct Option: (A)** $-12,500$

Distributive Property with Negative Terms:
Factor out 125:

$$125 \times [(-37) - 63] = 125 \times (-100) = -12,500$$

Q25 [MATHEMATICAL REASONING] **Correct Option: (A) Step 3**

Error Analysis:
The power of a power identity $(a^m)^n = a^{mn}$ is only universally valid when the base a is positive or when exponents are integers. Transforming $(-1)^{6/2}$ into $[(-1)^6]^{1/2}$ changes the order of operations and erroneously clears the negative sign inside the square root. Hence Step 3 is mathematically invalid.

Q26 [MATHEMATICAL REASONING] **Correct Option: (B)** -100

Pairwise Summation:
There are 200 numbers grouped into 100 pairs:

$$(1-2) = -1, \quad (3-4) = -1, \quad \ldots, \quad (199-200) = -1$$

Total sum:

$$100 \times (-1) = -100$$

Q27 [MATHEMATICAL REASONING] **Correct Option: (A)** -35

Extrema in Integer Sums:
List integer factor pairs (a, b) multiplying to -36:
- $1 + (-36) = -35$
- $2 + (-18) = -16$
- $3 + (-12) = -9$
- $4 + (-9) = -5$
- $6 + (-6) = 0$

The minimum sum is strictly $1 + (-36) = -35$.

Q28 [MATHEMATICAL REASONING] **Correct Option: (B)** 17

Maximizing Differences:
To maximize $(x - y)$, we must choose the maximum possible value for x and the minimum possible value for y:
- $\max(x) = 5$
- $\min(y) = -12$

$$\max(x - y) = 5 - (-12) = 5 + 12 = 17$$

Q29 [MATHEMATICAL REASONING] **Correct Option: (B)** -1

Integer Multiplicative Inverses:
An integer n is its own multiplicative inverse if $n \times n = 1 \implies n^2 = 1$. The only integer solutions are $n = 1$ and $n = -1$. Among the given options, -1 is the correct answer.

Q30 [MATHEMATICAL REASONING] **Correct Option: (B)** 50

Pairwise Difference:
Group into 50 pairs:

$$(100 - 99) + (98 - 97) + \cdots + (2 - 1) = \underbrace{1 + 1 + \cdots + 1}_{50 \text{ times}} = 50$$

Q31 [MATHEMATICAL REASONING] **Correct Option: (D)** 17

Linear Equation Solving:
1. Expand brackets:

$$-12x + 20 + 6 = -10x - 8 \implies -12x + 26 = -10x - 8$$

2. Transpose terms:

$$-12x + 10x = -8 - 26 \implies -2x = -34 \implies x = 17$$

Q32 [MATHEMATICAL REASONING] **Correct Option: (B)** 9

Triangle Inequality on Integers:
For any triangle with sides a, b, c:

$$|a - b| < c < a + b$$

Here, $|11 - 5| < x < 11 + 5 \implies 6 < x < 16$.
The integers are $x \in \{7, 8, 9, 10, 11, 12, 13, 14, 15\}$.
Count $= 15 - 7 + 1 = 9$ integers.

Q33 [MATHEMATICAL REASONING] **Correct Option: (D) Strictly positive**

Sign and Parity Analysis:
- Since $k < 0$, $(-k)$ is strictly positive $\implies (-k)^3$ is strictly positive.
- Since $k \neq 0$, k^4 (an even power) is strictly positive.
- Product of two positive numbers is strictly positive.

Q34 [MATHEMATICAL REASONING] **Correct Option: (B)** 9

Absolute Value Arithmetic:
1. $|-12 - (-7)| = |-12 + 7| = |-5| = 5$
2. $|5 - (-8)| = |5 + 8| = |13| = 13$
3. $|-9| = 9$
4. Combine:

$$5 + 13 - 9 = 9$$

Q35 [MATHEMATICAL REASONING] **Correct Option: (B)** -120

Product of Odd Count of Negative Numbers:
Count of negative factors is 5 (an odd number), so the product is negative:

$$-(2 \times 3 \times 4 \times 5 \times 1) = -120$$

Q36 [EVERYDAY MATHEMATICS] **Correct Option: (C)** $-\text{Rs } 3,000$

Bank Balance with Overdraft:

$$\text{Final Balance} = 4500 - (2800 + 3200 + 1500) = 4500 - 7500 = -\text{Rs } 3,000$$

(an overdraft of Rs 3,000).

Q37 [EVERYDAY MATHEMATICS] **Correct Option: (C)** $1,800$ metres

Vertical Separation Difference:

$$\text{Distance} = 1420 - (-380) = 1420 + 380 = 1,800 \text{ metres}$$

Q38 [EVERYDAY MATHEMATICS] **Correct Option: (A)** -8°C

Net Temperature Tracking:

$$\text{Midnight Temperature} = -4 + 9 - 13 = 5 - 13 = -8^\circ\text{C}$$

Q39 [EVERYDAY MATHEMATICS] **Correct Option: (B)** 52 marks

Olympiad Marking Calculation:

- Correct answers: $16 \times (+4) = 64$ marks.
- Incorrect answers: $(22 - 16) = 6$ questions $\implies 6 \times (-2) = -12$ marks.
- Total Score:

$$64 + (-12) = 52 \text{ marks}$$

Q40 [EVERYDAY MATHEMATICS] **Correct Option: (D)** -252 metres

Shaft Descent Position:

$$\text{Position} = +18 - (6 \times 45) = 18 - 270 = -252 \text{ metres}$$

Q41 [EVERYDAY MATHEMATICS] **Correct Option: (A) Net profit of** $\text{Rs } 20,000$

Net Financial Aggregation:

$$\text{Net} = (+45000) + (-28000) + (-35000) + (+52000) + (-14000)$$
$$= (45000 + 52000) - (28000 + 35000 + 14000) = 97000 - 77000 = +\text{Rs } 20,000$$

Q42 [EVERYDAY MATHEMATICS] **Correct Option: (A)** 275 metres

Relative Elevation Difference:

$$\text{Difference} = (-155) - (-430) = -155 + 430 = 275 \text{ metres}$$

Q43 [EVERYDAY MATHEMATICS] **Correct Option: (D)** -70 points

Net Stock Point Tracking:

$$\text{Net Change} = (-145) + 280 + (-95) + (-110) = 280 - (145 + 95 + 110) = 280 - 350 = -70 \text{ points}$$

Q44 [EVERYDAY MATHEMATICS] **Correct Option: (C) Floor** 16

Elevator Floor Tracking:

$$\text{Arrival Level} = -3 + 19 = 16 \implies \text{Floor } 16$$

Q45 [EVERYDAY MATHEMATICS] **Correct Option: (C)** -16°C

Cooling Rate Calculation:

1. Duration: $1 \text{ hour} = 60 \text{ minutes} = 12$ intervals of 5 minutes.
2. Total temperature drop: $12 \times 4^\circ\text{C} = 48^\circ\text{C}$.
3. Final temperature:

$$32 - 48 = -16^\circ\text{C}$$

Q46 [HOTS] **Correct Option: (A)** 9

Median Minimization of Absolute Distances:

For any sum $\sum |x - a_i|$, the minimum value occurs at the median of the given points $\{a_i\}$.

Here, the points are $2, 5, 11$. The median point is $x = 5$.

Evaluating at $x = 5$:

$$E(5) = |5 - 2| + |5 - 5| + |5 - 11| = 3 + 0 + 6 = 9$$

For any $x \neq 5$, $E(x) > 9$. Hence the minimum value is strictly 9.

Q47 [HOTS] **Correct Option: (C)** 12

Diophantine Integer Factoring (Simon's Favorite Factoring Trick):

Rearrange to group terms:

$$x(y + 3) - 5y = 33$$

Subtract 15 from both sides to factor:

$$x(y + 3) - 5(y + 3) = 33 - 15 = 18 \implies (x - 5)(y + 3) = 18$$

The number of integer solutions equals the number of integer factor pairs of 18.

Divisors of 18: $\pm1, \pm2, \pm3, \pm6, \pm9, \pm18$.

There are 6 positive divisors and 6 negative divisors, giving exactly 12 integer divisors.

Each divisor provides a unique integer pair (x, y). Hence there are exactly 12 pairs.

Q48 [HOTS] **Correct Option: (B)** 10

Parity and Product Invariants:

Let the numbers be $x_1, x_2, \ldots, x_{20}$ where $x_i \in \{+1, -1\}$.

The product of all 20 adjacent products is:

$$\prod_{i=1}^{20} p_i = \prod_{i=1}^{20} (x_i x_{i+1}) = x_1^2 x_2^2 \ldots x_{20}^2 = (+1)^{20} = +1$$

Since the product of all p_i is $+1$, the number of negative products (-1) must be EVEN.

If the sum of 20 terms (each ±1) equals 0, there must be equal numbers of $+1$ and -1:

$$\text{Count of } (+1) = 10, \quad \text{Count of } (-1) = 10$$

Since 10 is even, this is completely consistent. Thus, exactly 10 products equal -1.

Q49 [HOTS] **Correct Option: (D)** 77

Discrete Difference Optimization:

To maximize $(a - b) \times (c - d)$, we should make both differences as large positive numbers (or both as large negative numbers).

1. The maximum possible difference from the set is $6 - (-5) = 11$.

Let $a = 6, b = -5$.

2. The remaining available integers are $\{-3, -1, 2, 4\}$.
3. The maximum difference from the remaining set is $4 - (-3) = 7$.

Let $c = 4, d = -3$.

4. Product:

$$11 \times 7 = 77$$

Q50 [HOTS] **Correct Option: (D)** 51

Alternating Series Decomposition:

Group the first 100 terms into pairs of $(n - (n + 1))$:

$$S = (1 - 2) + (3 - 4) + \cdots + (99 - 100) + 101$$

There are 50 pairs of -1:

$$S = 50 \times (-1) + 101 = -50 + 101 = 51$$

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